

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 17:36:35

Training Start: 2025-05-07 16:56:50

Training End: 2025-05-07 17:31:01

Training Duration: 34.17 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.5949

MSE: 5.0796

RMSE: 2.2538

R2 Score: 0.1286

Multi-Step Forecast:

Forecast Steps: 40

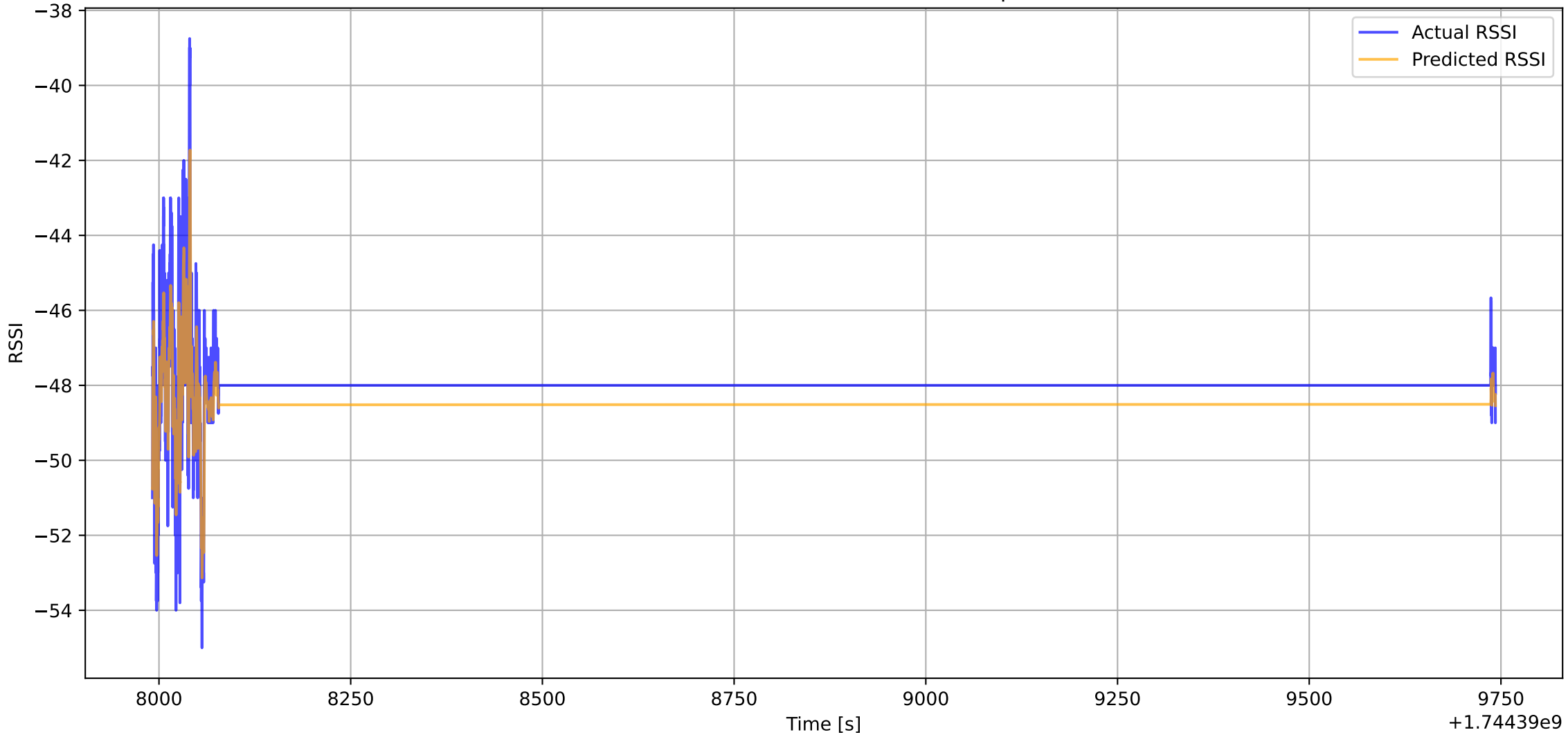
MAE: 0.0364

MSE: 0.0023

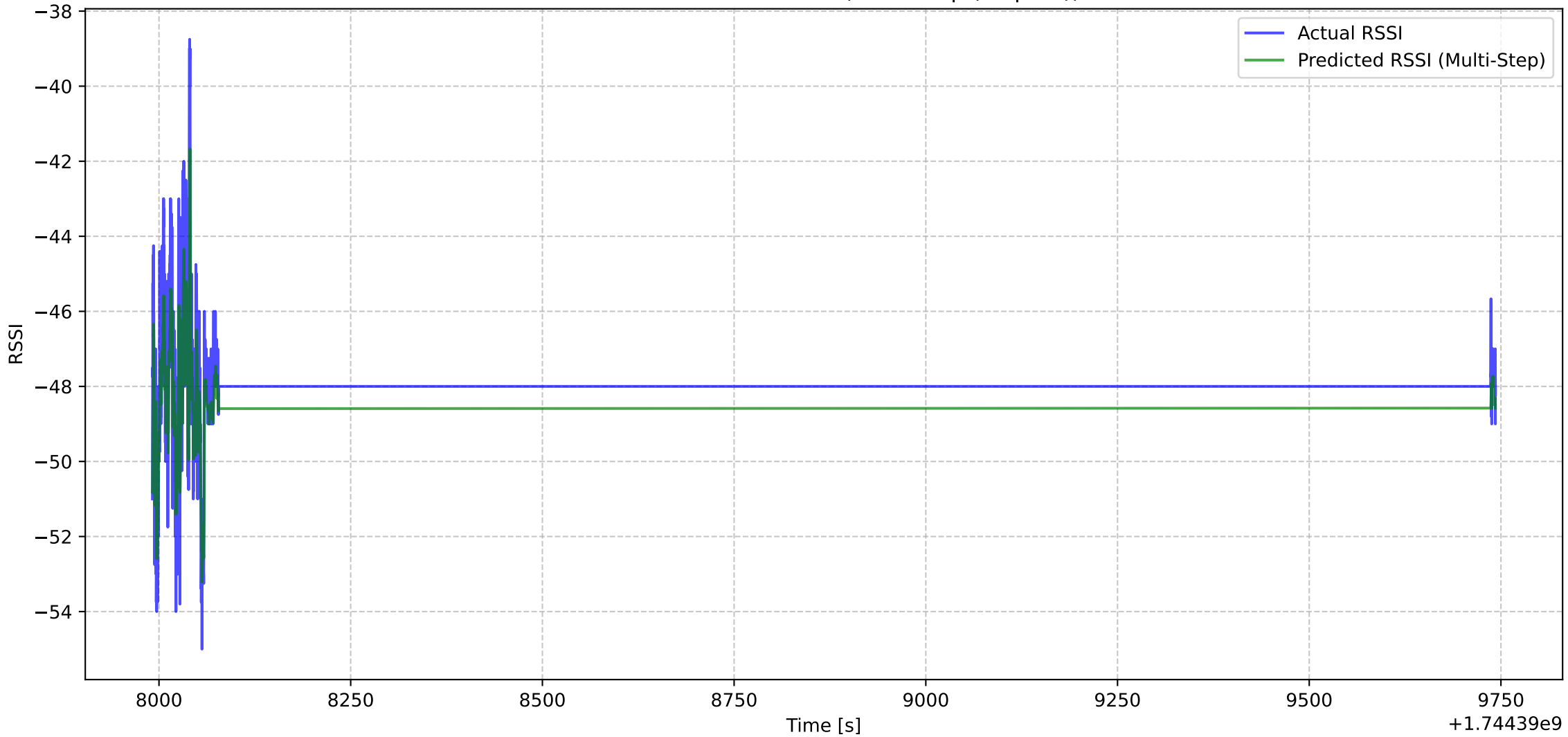
RMSE: 0.0479

R2 Score: 0.7105

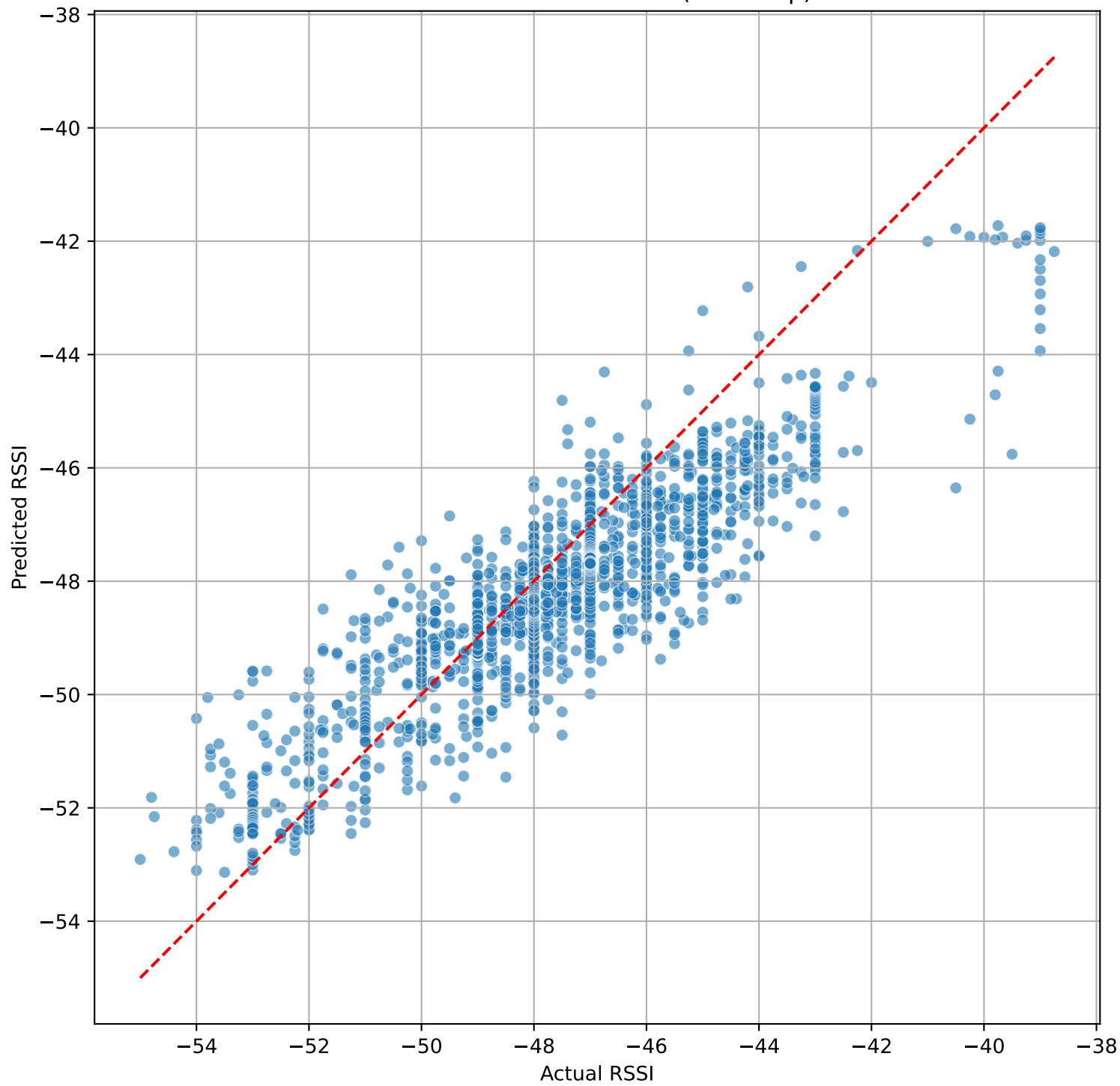
Actual vs Predicted RSSI (First Step)



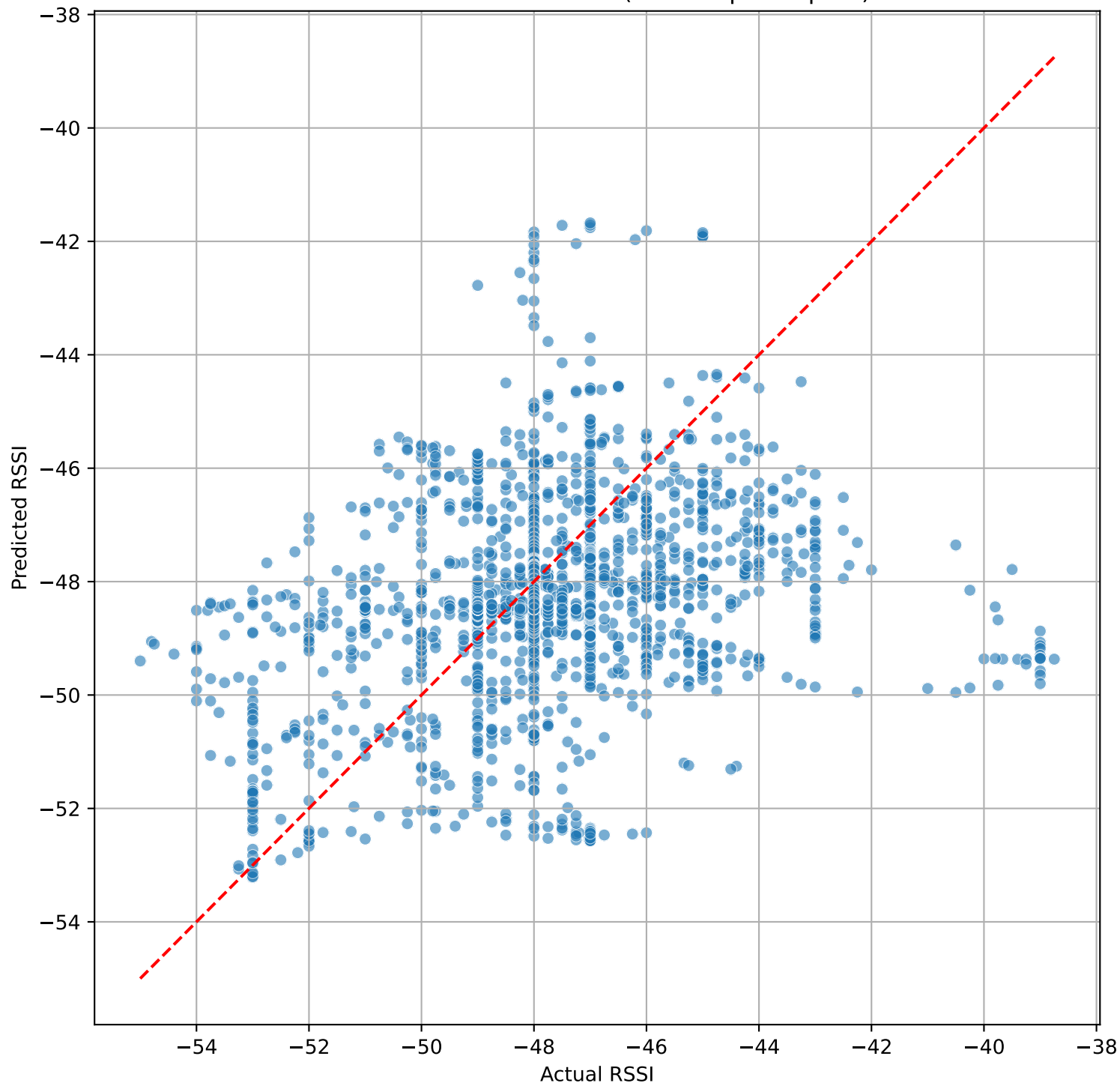
Actual vs Predicted RSSI (Multi-Step (step 40))



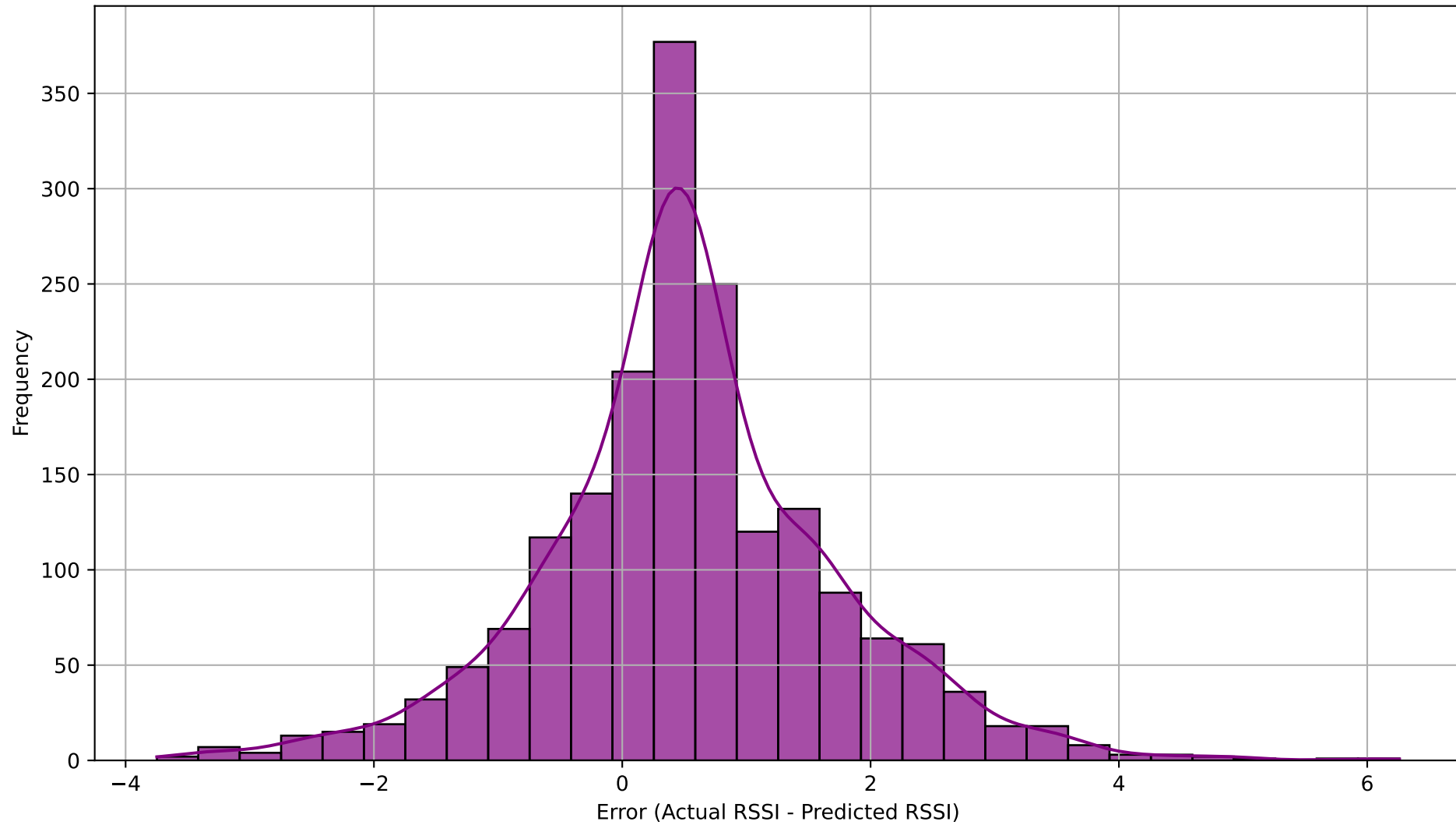
Actual vs Predicted RSSI (First Step)



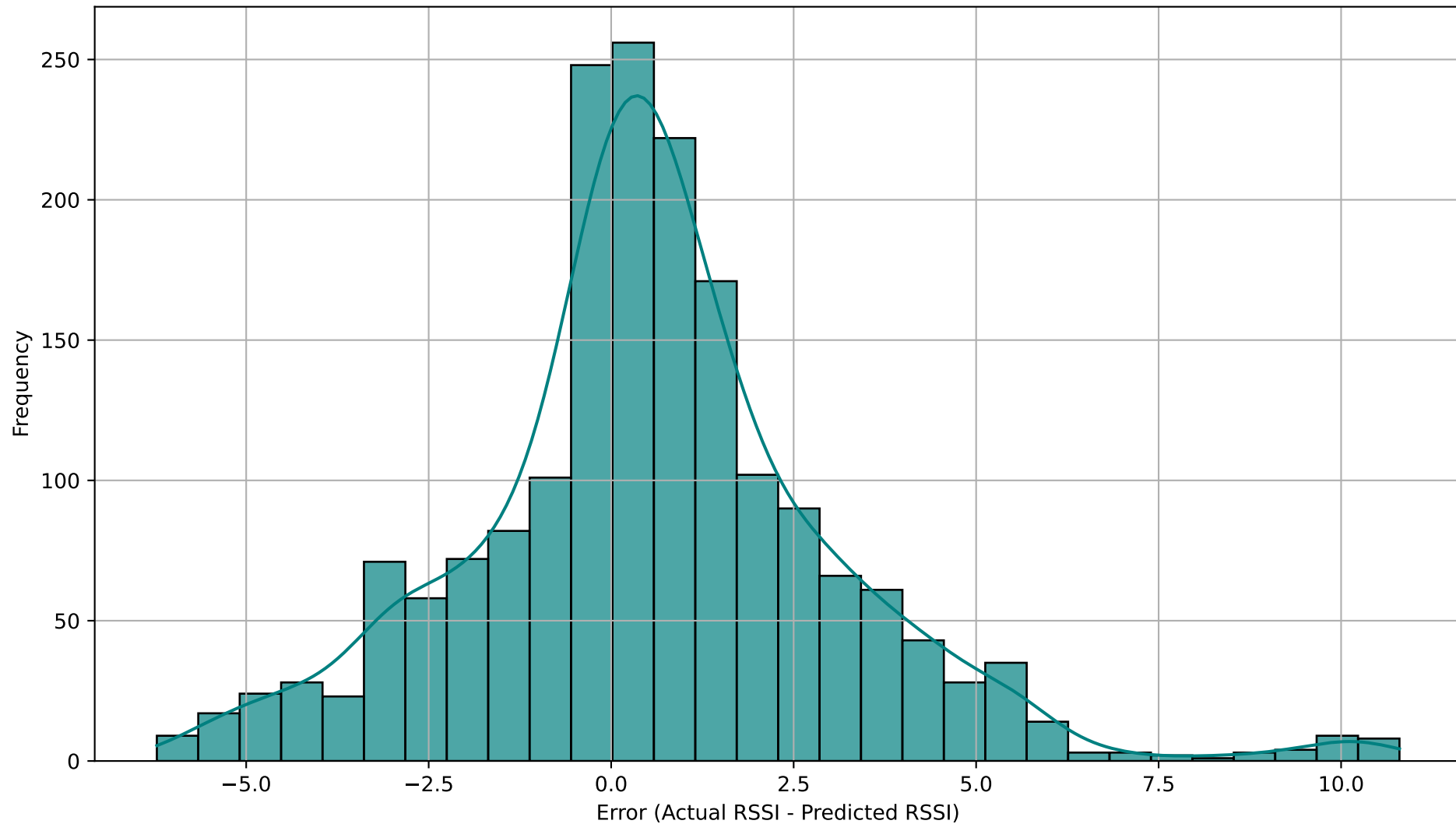
Actual vs Predicted RSSI (Last Step - step 40)



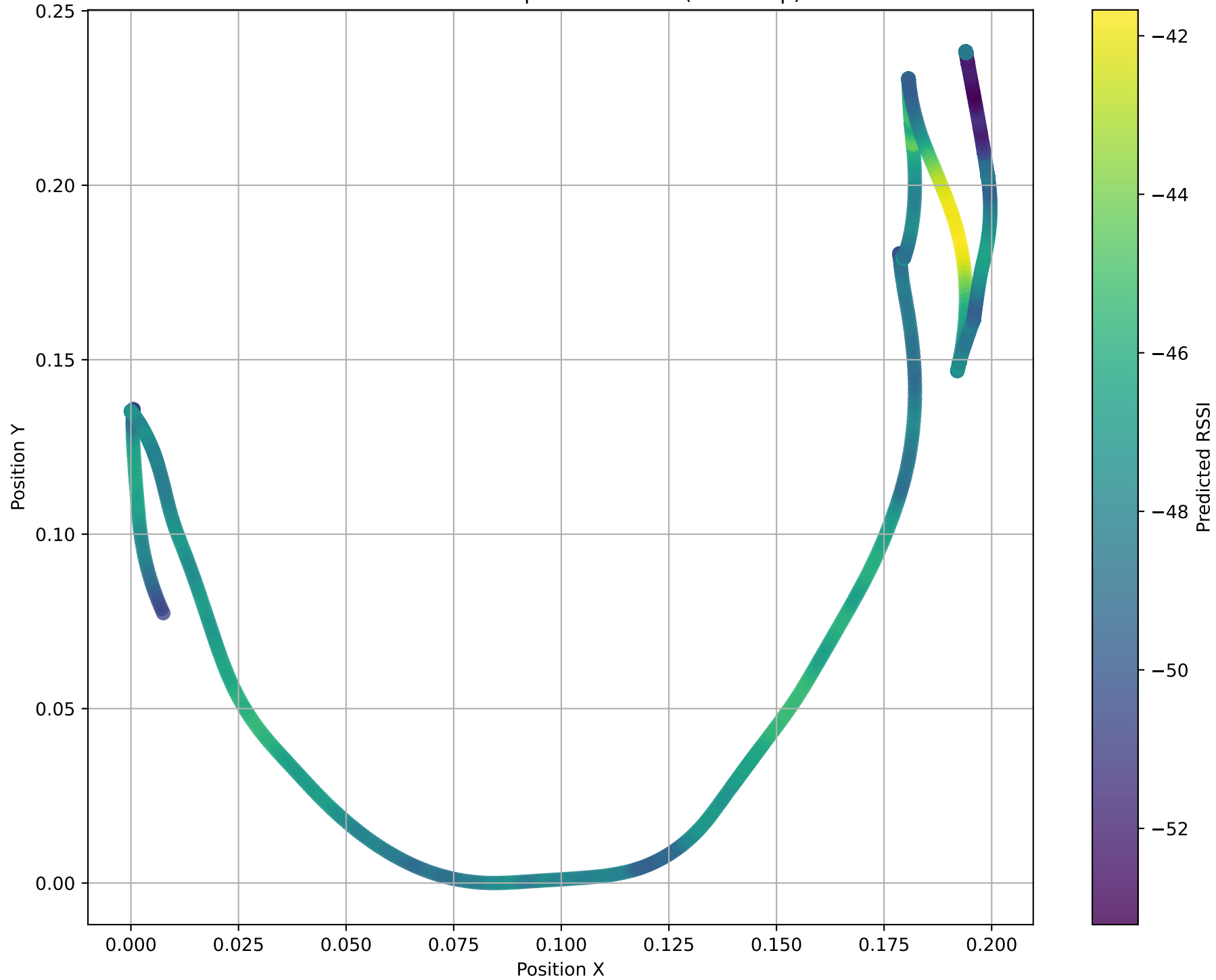
Distribution of Prediction Errors (First Step)



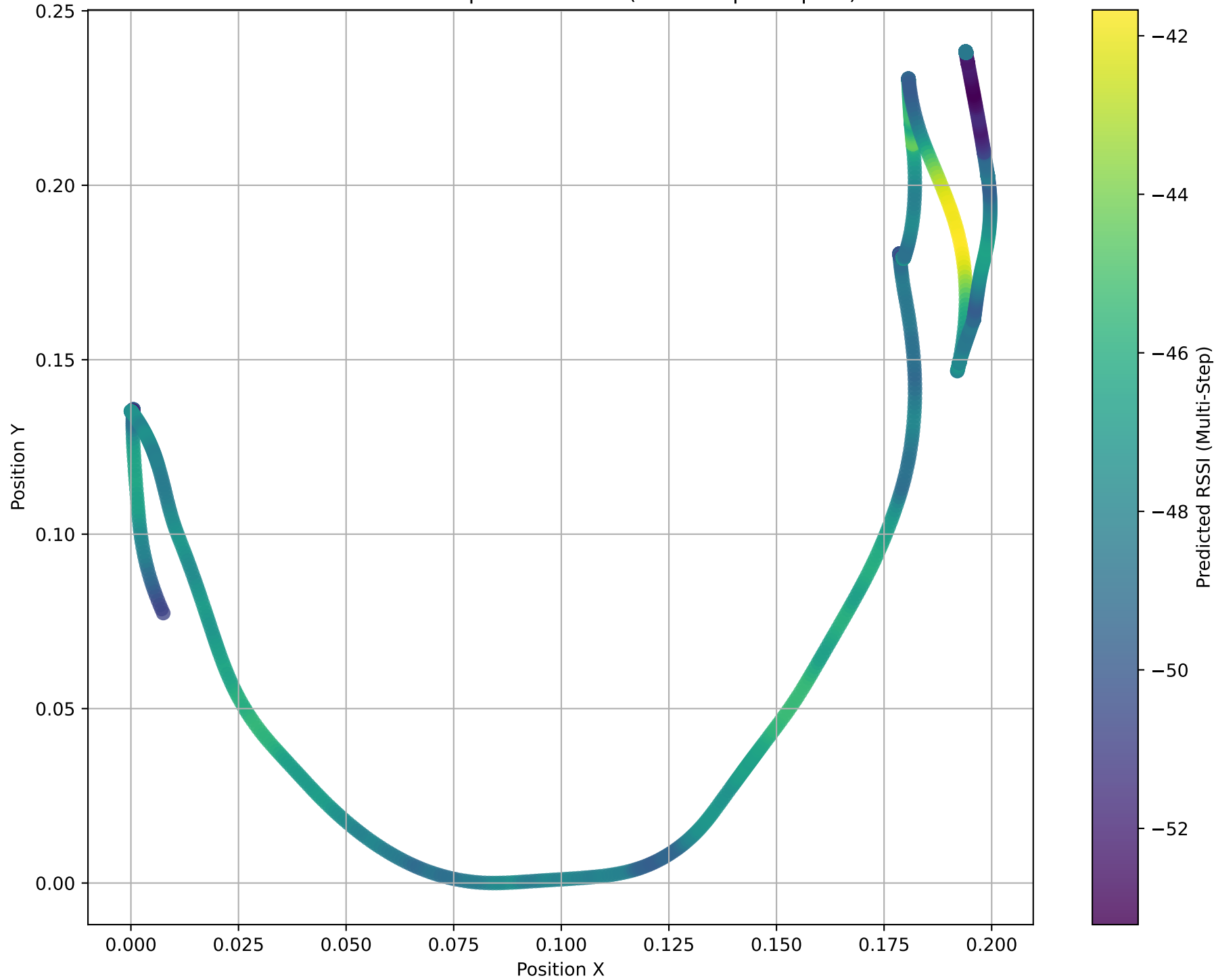
Distribution of Prediction Errors (Last Step - step 40)



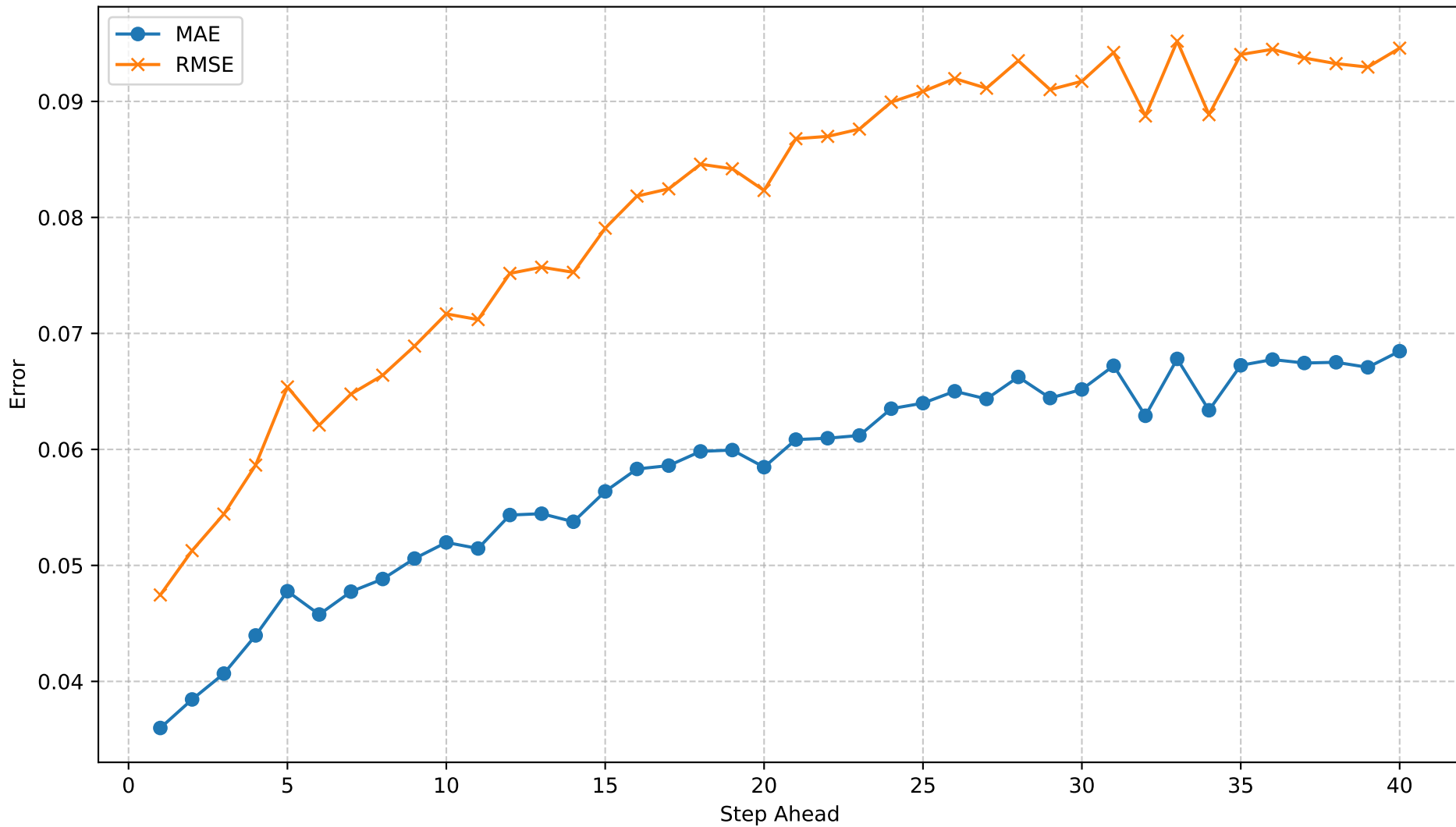
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25.parquet

Modelo: LSTM

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.05
- Mediana : 0.06 / 0.09
- 40º passo : 0.07 / 0.09

Crescimento do erro: 90.3% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 40 (0.07), 33 (0.07), 36 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.